One Stream, Many Windows: Soft Targets for Consolidating Fast and Slow Conversions

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Abstract. Conversion pipelines are split by speed: an upper-funnel onsite conversion arrives in minutes, a down-funnel offsite one in days. Each objective's *attribution window* — what counts as a conversion — is conventionally also its *join delay*, the age at which its training record is written, so every objective needs its own pipeline and is stale by its own window. We ask whether the two can be separated: can a long-window objective learn from records joined within 90 minutes?
Joining sooner records every in-flight conversion as a negative, and a single-visit stream, which trains each click at most once, cannot repair it later — yet most published corrections do exactly that: of 11 methods we reproduce, 8 write more than one record per click, up to 4.00. The gap must instead be closed at write time, in production by a training-only foundation model such as GEM, here by a 57,272,515-parameter teacher. On the Criteo Delayed Feedback Dataset, across five served capacities (1,890–9,580,477 parameters, 0.001–0.286 MFLOPs), three join delays and 10 seeds, we compare three targets: the matured label (Wait), the teacher's estimate alone (Distill-Only), and the observed label with only its negatives softened (Correct-Distill). Against a Wait baseline of NE 0.72–0.81, Distill-Only at one day lowers NE by 0.60% at the two smallest capacities but raises it by 0.51% at the three largest. Correct-Distill beats Distill-Only in all 30 configurations and Wait in 28, lowering NE by 0.48–2.68% on the seven-day task and 0.57% on average on the one-day task — 74% and 69% of what a label-peeking oracle recovers. The gain grows as the funnel slows and needs no larger teacher: a 6× larger teacher and a 4.7× more expensive one each move it by under 0.114 pp. Decoupling the two delays removes the main obstacle to training objectives with different attribution windows on one fresh stream.

Keywords: Conversion rate prediction · Delayed feedback · Online advertising · Knowledge distillation · Streaming training pipelines.

1 Introduction

Conversion pipelines are split by speed. An upper-funnel onsite conversion — a view, an add-to-cart — is observable within minutes of the click; a down-funnel offsite one, a purchase or an app install, arrives after a day or more. Each objective waits out its own *attribution window* before its training record exists, and conventionally that window is also its *join delay*: the age at which the

record is written. Equating the two forces every objective onto its own pipeline and makes each stale by exactly the width of the window it waited for.

A slow objective therefore trains on week-old data, and consolidating several objectives into one model inherits the worst of every window (Sect. 6). We ask whether the two delays can be separated: can an objective accountable to a seven-day window be trained from records joined within 90 minutes?

Joining sooner is not free. A record written at age o records every conversion still in flight as a negative, and the deficit is large — at $o = 5$ minutes only 0.355 of a one-day window's positives have arrived, and 0.265 of a seven-day window's. A substantial literature corrects this bias [1,11,10,2,3], but most corrections repair the record after the fact — duplicating the event when its outcome lands, relabelling it, adding heads at further horizons — and a pipeline that trains each click at most once can do none of these. We make that restriction explicit as a *single-visit* constraint and measure who obeys it: of 11 published methods reproduced under one protocol, 8 write more than one record per distinct click, up to 4.00 (Sect. 2).

What is left is a choice of target. If the record must be written at o and carry one frozen number, the gap between what is observed and what the objective means has to be closed at write time by an estimate. In production that estimate is naturally supplied by a large model trained across objectives on matured outcomes — a foundation model such as GEM [7] — attractive precisely because it already exists and is not refitted per objective. Here we stand in for it with a 57,272,515-parameter training-only teacher, and compare three targets across five served capacities, three join delays and two attribution windows. The correction works, the gain grows as the funnel slows, it needs no larger teacher — and it is conditional on the record keeping part of its hard label.

2 Setting and the Single-Visit Constraint

A click i occurs at C_i with features x_i and, if it converts, after a delay $D_i > 0$ ($D_i = \infty$ otherwise). The *attribution window* v belongs to the objective and fixes the target $Y_i^{(v)} = \mathbf{1}[D_i \le v]$, knowable only at $C_i + v$; the *join delay* o belongs to the pipeline and decides when the record is written, carrying

$$Y_i^{(o)} \;=\; \mathbf{1}[D_i \le o] \;\le\; Y_i^{(v)}. \tag{1}$$

Setting $o = v$ is the segregated status quo; $o \ll v$ is what we want, and the inequality is its price. Write $F(o) = P(D \le o \mid D \le v)$ for the share of the target's positives visible at write time. Training runs in hourly rounds — consume the records released since the last cut-off, then score every click of the next hour once, retrospectively against $Y^{(v)}$ — so every arm is judged on the business label whatever it trained on, and update-then-predict keeps the evaluation causal.

Two rules govern what a deployable system may do. **Rule 1 (causality):** a record's label must be computable when the record is written and must not change afterwards. Labelling a record $Y^{(v)}$ while writing it at $C_i + o$ uses an

outcome that has not happened — an upper reference (ORACLE), not a system. **Rule 2 (single-visit):** with n_i the number of times click i enters the served model's loss over its lifetime, $\max_i n_i \leq 1$. Without it shortening o is trivial — write early, write again later. The criterion is the maximum rather than the mean, since it is the *existence* of a second write that obliges the pipeline to support one, and it governs the *served* stream: charging training-only auxiliaries too would eliminate FSIW, ULC and our own arms alike.

Reproducing 11 published methods from scratch under one protocol and counting the records each one's served model consumes per distinct click, 8 write more than one.[1] Records per click, at $v = 1\,\mathrm{d}$: FTP 4.00, MISS 3.05, IF-DFM 2.79, PI 2.00, DDFM 1.89, DFM 1.33, DEFUSE 1.03, ES-DFM 1.03 — and IF-DFM reaches 97 writes for a single click, its mechanism being an update applied whenever a label reverses. The plain re-ingestion baseline [5,4] measures 1.03 and fails the same criterion. Duplication is a property of emission logic, not of the model: the audit gives identical counts at the smallest and largest capacities. Only FSIW, TWICE, ULC are single-visit; of these FSIW corrects the *loss* and TWICE adds a head on the conversion clock, both leaving the target untouched, while ULC alone changes the target, which makes it our closest comparator.

3 What Fills the Gap

Shortening o leaves a gap: conversions arriving between o and v are recorded as negatives, and something must stand in for them in the record's single frozen target. We study four answers, identical in pipeline, schedule, features and optimiser, differing only in the target $\tilde{y}_i$ and the age Δ at which the record is written.

FRESH ($\tilde{y}_i = Y_i^{(o)}$ at $\Delta = o$) fills the gap with nothing. WAIT ($\tilde{y}_i = Y_i^{(v)}$ at $\Delta = v$) avoids having one by deferring the write: correct label, v-old data. It is the segregated status quo and our baseline throughout — a practitioner who refuses both to duplicate records and to train on wrong labels is already running it. CORRECT-DISTILL writes at $\Delta = o$ and softens only the observed negatives,

$$\tilde{y}_i \;=\; Y_i^{(o)} + (1 - Y_i^{(o)})\, w^T(x_i), \qquad w^T(x) \approx P\Big(Y^{(v)}{=}1 \,\Big|\, Y^{(o)}{=}0,\, x\Big), \tag{2}$$

so an observed positive stays exactly 1 and only the uncertain mass moves. This is the functional form of ULC [8]; what differs is the estimator, so a comparison between them isolates that and nothing else. DISTILL-ONLY goes further and regresses on $\hat{p}_v(x) \approx P(Y^{(v)}{=}1 \mid x)$ alone: the record carries features and a number, and no observation at all.

[1] The denominator must be *distinct clicks touched*: dividing records consumed by records released conflates duplication with traffic variation. We also score only clicks whose whole revisit horizon lies inside the stream, since a shorter window cannot see the revisit.

The teacher. A separate model trained on the *matured* stream — a record enters it at $C_i + v$ carrying $Y_i^{(v)}$ — with $\hat{p}_v$ and w^T as read-outs on one trunk. It is *evaluated* at the write instant: at $C_i + o$ the student queries it and freezes the number into the record, so Rule 1 holds even though the target is soft: softness is another way of satisfying causality, not an exemption from it. The construction is distillation from privileged information [9]: the teacher sees the matured label, exactly what the student cannot.

Teacher capacity is 57,272,515 parameters (1.593 MFLOPs). Being training-only it costs nothing at serving time, but it does mean *served* capacity is matched across arms while total training capacity is not — which is why teacher size is an experimental axis rather than a fixed detail. Nor need the teacher be live: ours keeps learning from the matured stream, making it a service, whereas frozen after causal initialisation it is a build-time artifact. We run both. ORACLE, writing at $\Delta = o$ but labelling the record $Y_i^{(v)}$, violates Rule 1 and bounds what any correction could recover.

4 Experimental Setup

Data and protocol. The public Criteo conversion logs released with [1]. Clicks span 60 days; days $[0, 30)$ initialise the model causally and days $[30, 60)$ are the streaming period, divided into 719 hourly rounds giving 8,196,131 scored clicks at a base conversion rate of 0.1361. Learning rates are selected per arm and per cell from a seven-point grid on a held-out slice of the initialisation period, before any streaming evaluation, and pinned across seeds. The teacher's rate is fixed and decoupled from the student's, so the student's sweep tunes one model rather than two.

Axes. Two *attribution windows*, $v = 1\,\mathrm{d}$ and $7\,\mathrm{d}$, standing for a fast and a slow funnel; the contrast between them is the paper's point, so we never average across them. Three *join delays*, $o \in \{5, 30, 90\}$ minutes. Five *served capacities* from 1,890 to 9,580,477 parameters and 0.001–0.286 MFLOPs per inference — the two do not track, the smallest rungs differing $5.2\times$ in parameters at identical compute, so we report both. 10 seeds on the three targets and 5 elsewhere: 2,050 runs over 14 arms, alongside the 4,850-run screen of Sect. 2.

Metric and statistics. Normalised entropy against $Y^{(v)}$, lower better, pooled over all scored clicks rather than averaged over rounds — a per-round average normalises each hour by its own base rate and discards diurnal variation a model can legitimately predict. We report the relative reduction against WAIT within a cell, computed per seed and then averaged. Seeds differ only in initialisation and minibatch order, so seed i of two arms is a matched pair: paired t-test within each cell, Holm–Bonferroni across the 15 cells of a contrast, counting only cells whose *gain* is positive and survives. Dispersion is small away from FRESH (median NE sd 0.00107, maximum 0.00520); FRESH reaches 0.0490 at $o = 5$ minutes, which is a result rather than a nuisance. Seeds measure initialisation variance on one data origin: precision, not generalisation.

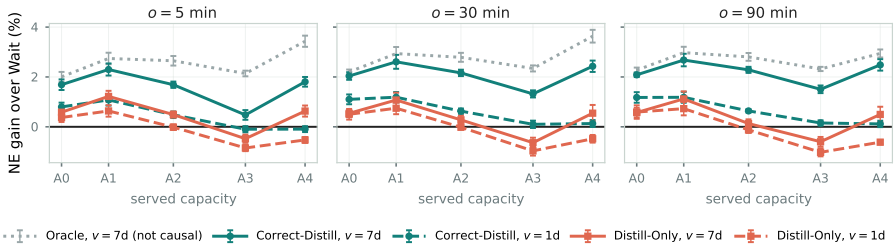


Fig. 1. NE gain over Wait against served capacity, one panel per join delay; above the line beats the segregated status quo. Colour is the target, dash the attribution window, and Oracle is drawn at $v = 7\,\mathrm{d}$ to bound the panel. Bars are the standard deviation of the per-seed differences.

5 Results

Throughout, the reported quantity is the relative NE reduction against Wait — the segregated status quo that joins on the attribution window — so positive means the short join is better.

The join window can be collapsed. Writing the record 2,016 times sooner than the seven-day window it answers to, and letting the teacher stand in for the conversions not yet seen, lowers NE in all 15 seven-day cells (15 significant under Holm–Bonferroni) by 0.48 to 2.68%, mean 1.97%. At one day the same substitution gains 0.57% on average, positive in 13 of 15 cells and significant in 10. The two windows differ by a factor of three, and the direction is the useful part: *the slower the funnel, the more collapsing its join window is worth.* That inverts the intuition — a longer window censors more of the label, so the teacher has more to supply — because what dominates is the cost of the alternative, and no label accuracy compensates for training on week-old data.

Both ends of the interval are measured. Oracle, writing at o but labelling the record $Y^{(v)}$, gains 2.68% and 0.83%, so Correct-Distill recovers 74% and 69% of what any causal correction could; Fresh, the same join uncorrected, loses 20.1% and 11.8%. The gap the teacher fills is an order of magnitude larger than the gains above.

The condition: keep part of the hard label. The collapse above used the hybrid target; the fully soft one decides how far the result can be pushed. At $v = 1\,\mathrm{d}$, Distill-Only beats Wait at the two smallest capacities by 0.60% and *loses* at the three largest by 0.51%, crossing over between A1 and A2 at every join delay (Fig. 1). A purely soft target plausibly acts as a regulariser, a variance reduction a small model can use and a larger one no longer needs [6]. This cuts against convenience: a record with no hard label is the most portable object a consolidated pipeline could ask for, and it is exactly the one that fails at the capacities worth serving. At $v = 7\,\mathrm{d}$ the crossover is weaker and non-monotone

(0.10% across the three largest rungs), so we state the rule for the one-day window and do not extrapolate it. CORRECT-DISTILL beats DISTILL-ONLY in all 30 cells, though not WAIT: it is below WAIT in 2 of 30 cells, both at $v = 1\,\mathrm{d}$ and $o = 5$ minutes, at rungs A3, A4, the worse by 0.097%, and neither survives Holm–Bonferroni. The location is coherent: the fastest join at the fast window is where the teacher has least to add, 0.355 of the positives being already visible.

The teacher need not be large, costly or live. Substituting a 9,456,897-parameter teacher for the 57,272,515-parameter one, a 6× parameter and 41× compute step, moves the gain by -0.043 pp for CORRECT-DISTILL and +0.114 pp for DISTILL-ONLY over 30 cells. Holding parameters fixed and spending 4.7× more teacher compute is equally inert (2.11% against 2.11% at seven days). The teacher's rate is verified, not inherited: the default beats a 10× smaller one everywhere (1.75%) and ties a 2× larger one (2.12%), so the null is a saturated teacher rather than an untuned one. Freezing it after causal initialisation costs 0.08 and 0.27 pp, and the frozen arm still beats WAIT in 27 of 30 cells: a shipped model suffices.

CORRECT-DISTILL and ULC [8] share Eq. (2) and differ only in how the corrector is estimated; on the matched grid they are near-indistinguishable (mean NE difference +0.00049, 14 of 15 cells not significant at $v = 1\,\mathrm{d}$). The offline teacher does not beat the online corrector on accuracy; the case for it is architectural (Sect. 6). Relaxing Rule 2, by letting the teacher read the fresh stream at 2.08 visits per click, gains 1.19% at one day against 0.63% and nothing at seven, so single-visit is near-free on the slow funnel.

6 What This Means for Consolidation

Consolidating ranking objectives into one model is an active industrial direction [7], and the obstacle is a pipeline one: under WAIT objective k's record exists only at $C_i + v_k$, so a shared append-only stream must either write K records, violating Rule 2 K times over, or delay everything to $\max_k v_k$. Our result removes the dilemma: one stream at a single short o can carry $\tilde{y}_{i,k} = Y_{i,k}^{(o)} + (1 - Y_{i,k}^{(o)})\, w_k^T(x_i)$ for every objective at once, because w_k^T is a function the pipeline calls rather than an outcome it awaits — and doing so is an improvement, most of all for the slow objectives that forced the segregation.

ULC evaluates at write time too and reaches the same accuracy, so the choice is architectural: it fits each w_k inside the served model's training loop, turning K objectives into K auxiliaries on one optimiser, whereas a teacher puts each w_k upstream with a number on a record as its only interface. Sect. 5 says what that upstream model must be — not large, not costly, not live — so consolidation adds K build-time artifacts rather than K services. That is the regime in which a shared cross-objective foundation model is attractive: not because it is big, but because it already exists, already sees matured outcomes for every objective, and can be evaluated once per record and then forgotten.

What this does not establish. We ran one objective at a time, so there is no evidence on interference between objectives sharing a trunk — the central risk, and what a multi-task study must measure. Our teacher is a stand-in for the production foundation model the argument invokes, not an instance of one, and one corpus with one split measures precision rather than generalisation.

Conclusion. The join window a pipeline waits for can be decoupled from the attribution window it answers to: shortened by three orders of magnitude, at better accuracy, recovering 74% of what a label-peeking oracle could at seven days, and most valuable where the funnel is slowest — provided the record keeps part of its hard label. The fully label-free target, the most convenient to consolidate, is the one that fails. The experiment this makes worth running is K heads, K windows, one stream, teachers upstream.

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